



Method for Uric Acid as an Antioxidant Treatment for LRRK2-associated Inherited Parkinson's Disease

Ning Xia and Rachit Bakshi

MassGeneral Institute for Neurodegenerative Disease
Massachusetts General Hospital, Boston, MA 02129

PPMI Project ID: 137

Summary

Through this project, we seek to utilize the iPSC cells from PPMI repository which are derived from participants who have idiopathic PD, genetic PD and healthy volunteers and differentiate them to dopaminergic neurons to use as a model system to test urate's neuroprotective potential. Since we have previously documented that urate can produce much of its neuroprotective effect indirectly via primary astrocytes, we will culture the iPSC-derived neurons with conditioned medium derived from human astrocytes. This model also could be useful to study functional interactions between patient iPSC-derived neurons and releasable human astrocyte factors in the presence of PD toxins $\pm$ urate treatment. We will assess whether urate treatment prevent or slow progression of PD-related phenotypic alterations in iPSC-derived DA neurons such as cell death via oxidative and proteasomal stress, mitochondrial deficits, and neuroinflammation.

Methods

Induced pluripotent stem cell (iPSC) maintenance and dopaminergic neurons differentiation.





iPSC cells are cultured on vitronectin coated 35mm dishes in StemFlex medium (GIBCO Thermo Fisher). RevitaCell™ Supplement was used as a post-thaw recovery solution to improve cell viability and in combination with StemFlex Medium to enhance single cell passaging before differentiation. Cells were maintained at 37 C with 5% CO₂. PSC Dopaminergic Neuron Differentiation Kit (GIBCO) was used to differentiate iPSC cells. In brief, on day -1, prepare a single cell suspension of iPSCs and seed each well of the vitronectin-coated 6 well plates with 4.0×10^4 viable cells/cm² in complete StemFlex Medium + 10 μM ROCK inhibitor (Y27632) (Sigma-Aldrich, Cat. No. Y0503). Incubate the cells overnight at 37°C in a humidified atmosphere of 5% CO₂. After 24 hours, replace the medium with 2mL pre-warmed complete specification medium and refresh every other day with specification medium until day 10. Expand floor plate progenitor cells (FP cells) from day 10. Passage PF cells on day 10, day 12 on laminin coated 6-well plates in complete Floor Plate Cell Expansion Medium. On day 16, harvest the FP cultures for suspension culture to form spheres. Resuspend the FPp2 cells to 1.0×10^6 viable cells/mL in complete expansion medium + 5 μM ROCK inhibitor (Y27632), Transfer cell suspension to Nunclon™ Sphera™ cell culture 6-well plates, complete medium change on day 17 and 19 for sphere culture. Dissociate spheres on day 21, resuspend the cell pellet to a single cell suspension in complete maturation medium + 5 μM ROCK inhibitor (Y27632) and seed in double-coated with 100 μg/mL poly-D-lysine and 15 μg/mL laminin dishes at a density of 2.0×10^5 cells/cm². On day 22 of differentiation, add the same volume of fresh complete maturation medium (without the ROCK inhibitor) as the existing culture





volume, every 2–3 days thereafter, aspirate half of the spent medium and replace it with fresh complete maturation medium until day 35.

Conditioned Media Experiments

Human astrocytes were purchased from ScienCell labs (<https://www.sciencellonline.com/human-astrocytes.html>). Astrocyte cultures reached confluence after 7–10 days in vitro. Urate was dissolved in DMEM as 20X concentrated stocks. Enriched astroglial cultures were treated with 100 μ M urate or vehicle. Twenty-four hours later conditioned media (CM) were collected and filtered through a 0.2 μ m membrane to remove cellular debris and immediately used for experiments. All reagents for cell culture were obtained from Life Technologies.

Treatment Groups

The differentiated dopaminergic neurons were incubated with CM from urate (UCM) or vehicle-treated astrocytes (CCM) for 24h before addition of 200 μ M H_2O_2 for another 24h. The other two groups were Control (with no CM and H_2O_2) and H_2O_2 only.

Immunofluorescence assay

On day 21, seed the floor progenitor cells on PDL and laminin double coated 8-um chamber slides. Differentiate cells for 2 weeks. Move the culture media and rinse with warm PBS. Cover cells with warm 4% PFA and fix for 20min at room temperature. Rinse three times in PBS for 5 min each. Block and Permeabilization





for 30min at room temperature using blocking buffer: 5%BSA and 0.3%Triton X-100 diluted in PBS. Dilute primary antibody (TH anti-rabbit polyclonal antibody, ENZO. BML-SA497-0100) 1:500 in 1/3 blocking buffer, incubate 2 hours at room temperature. Rinse three times in PBS for 5 min each. Incubate with secondary antibody (Goat-anti Rabbit IgG, 568), diluted at 1:250 in PBS, for 2 hours at room temperature in dark. Rinse twice in PBS for 5 min each. Incubate with DAPI (1ug/ml) for 10min. Rinse twice and cover with PBS. Take images using OLYMPUS IX70 microscope.

RNA Isolation and Quantitative Real-Time PCR

RNA was extracted from astrocyte cultures by TRIzol (GIBCO/BRL) extraction. RNA quality was determined by spectrophotometry and by visual inspection of electropherograms using the RNA 6000 NanoChip Kit on the Agilent 2100 Bioanalyzer (Agilent Technologies). cDNA synthesis was performed using Superscript@ VILO cDNA synthesis kit (Invitrogen). For quantitative gene expression analysis, SYBR green primers and probes (Applied Biosystems) were used. The specificity of each PCR product was confirmed by melting dissociation curve (T_m) analysis. The comparative threshold cycle method was used for quantitative analysis. Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) and RPL13 ribosomal RNA were used as RNA loading controls. Equal amplification efficiencies were confirmed for target and reference genes.

Western Blot Analysis





Western blot analysis was performed using lysates from differentiated dopaminergic neurons. The blots were probed with antibodies against Tyrosine Hydroxylase (Invitrogen, MA1-24654), alpha-synuclein (Abcam, ab138501) and β -actin (cell signaling technology, 4970) using the dilutions recommended in the product datasheet. Band densities were analyzed with Image Studio software.

About the Authors

This document was prepared by Ning Xia and Rachit Bakshi in the Department of Neurology of Massachusetts General Hospital, Boston, MA 02129. For more information please contact Rachit Bakshi at (617) 724-2575 or by email at RBAKSHI1@mgh.harvard.edu.

Notice: This document is presented by the author(s) as a service to PPMI data users. However, users should be aware that no formal review process has vetted this document and that PPMI cannot guarantee the accuracy or utility of this document.

